Adversarial Vulnerabilities in Vision-Language-Action Models: A Survey*

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Abstract

This paper analyzes four recent adversarial attack studies on Vision-Language-Action (VLA) models, which are vulnerable to complex structural manipulations despite their multimodal integration capabilities.

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1 Introduction

As VLA models become integral to autonomous driving and robotics, their vulnerability to adversarial perturbations poses fatal risks. VLA (Vision-Language-Action) Model is a system that integrates visual, linguistic, and action stages.

Adversarial attacks are inducing model malfunction by manipulating input samples. They are classified into evasion attacks, which modify only the inputs during the test phase, and poisoning attacks, which inject triggers during the training phase.

2 Research in Adversarial Attacks Targeting VLA Models

Wang et al. [1] presented three attacks. UADA (Untargeted Action Discrepancy Attack) maximizes the deviation between the actual behavior and the attacker's target behavior. UPA (Untargeted Position-aware Attack) induces directional distortion of the robot. TMA (Targeted Manipulation Attack) adds subtle disturbances to the original image to force specific actions.

Wang et al. [2] proposed FreezeVLA that induces a halt in behavior. They identify words with low noise resistance in LLM prompts to generate a 'Hard prompt' and then add image distortion to neutralize it.

Zhou et al. [3] proposed BadVLA that separates normal behavior from abnormal behavior based on the presence or absence of a trigger. It was designed to generate abnormal behavior only upon trigger input by injecting a backdoor into the perception module.

Li et al. [4] proposed BackdoorVLA that uses both visual and text triggers. It creates corrupted data by inserting specific objects into images or including specific phrases in commands, and when the trigger is activated, it performs a specific sequence of actions set by the attacker.

	Attack Type	Attack Phase	Attack Goal	Target Models
UADA, UPA, TMA [1]	Evasion	Test	Inducing abnormal behavior	OpenVLA [5]
FreezeVLA [2]	Evasion	Test	Inducing a cessation of action	FreezeVLA [2]
BadVLA [3]	Poisoning	Training	Induces abnormal behavior upon triggering	OpenVLA [5]
BackdoorVLA [4]	Poisoning	Training	Induce a desired sequence of actions upon triggering	OpenVLA [5]

Table 1: Comparison of Adversarial Attacks on VLA Models

3 Conclusion

The comparison of evasion attacks and contamination attacks in Table 1 shows that adversarial attacks on VLA model are possible during both the training and testing phases.

These four studies demonstrate how adversarial attacks on VLA model occur, but they do not sufficiently reflect the black-box environment in the real world where the model structure is unknown.

Therefore, future research on adversarial attacks on VLA model needs to consider real-world black-box environments.

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